

Version Control (Git & GitHub)

Your safety net — save your work, back it up, and undo mistakes. Claude Code handles all the commands.

The Six Concepts

Concept	What It Means	Analogy
Repository (repo)	A project folder tracked by Git	Your saved project
Commit	A saved checkpoint of your code	Hitting "save" in a video game
Push	Upload your commits to GitHub	Backing up your saves to the cloud
Pull	Download the latest version from GitHub	Syncing a cloud save to your device
Clone	Copy a GitHub repo to your computer	Starting a cloud save on a new device
Revert	Go back to a previous commit	Loading an old save

Step 1: Create a GitHub Repository

1. Go to **github.com** (sign in if needed)
 2. Click the **+** icon in the top right → **New repository**
 3. Name it to match your project (e.g., `my-first-project`)
 4. Set visibility to **Private**
 5. Do NOT check any initialization boxes (no README, no .gitignore)
 6. Click **Create repository**
 7. Copy the repo URL from the setup page (you'll need it next)
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Step 2: Connect Your Project to GitHub

Copy-paste this into Claude Code (replace the URL with yours):

```
Initialize a Git repository in this project and connect it
to my GitHub repo at [PASTE YOUR REPO URL HERE].
```

Then create your first commit:

```
Create a simple README.md that says "My First Project"
and commit it. Then push it to GitHub.
```

Go to your GitHub repo page and refresh — you should see your files.

First-time authentication: The first time you push, GitHub will ask you to sign in via a browser popup. Follow the prompts — this only happens once.

Step 3: The Commit-Push Cycle

This is the habit you'll use after every milestone:

```
Commit the changes and push to GitHub.
```

Good commit messages describe what changed:

- "Added hero section and navigation"
- "Fixed pricing calculator logic"
- "Completed landing page layout"
- "Added form validation"

When to Commit

Commit after every meaningful milestone — not after every tiny change, but regularly enough that you always have a recent save point.

Just finished...	Commit?
A feature or section	Yes
Fixing a bug	Yes

Just finished...	Commit?
A phase in your plan	Yes
A quick typo fix	Optional
Work before a break	Yes — always save before walking away
Before making big changes	Yes — create a save point to fall back to

Undoing Mistakes

Go back to the last commit (undo recent changes):

```
Revert to the last commit.
```

See what's changed since your last commit:

```
Show me what files have changed since the last commit.
```

Go back to a specific point in history:

```
Show me my recent commit history so I can pick  
a point to go back to.
```

Pull — Get the Latest from GitHub

Pull is the opposite of push. Use it when GitHub has a newer version than your computer.

When you'll use pull:

- You worked on the project on a different computer and pushed from there
- A teammate made changes and pushed them
- You edited a file directly on GitHub in the browser

```
Pull the latest changes from GitHub.
```

Clone — Copy an Existing Repo

Use clone to download an existing GitHub project to your computer for the first time.

When you'll use clone:

- Starting on a new computer with a project you already have on GitHub
- Picking up a project someone else built
- Working on a shared project with a team

```
Clone the GitHub repo at [PASTE REPO URL] into my Projects folder.
```

After cloning, open the new folder in VS Code and you're ready to continue working.

Viewing Your History on GitHub

On your GitHub repo page:

- **Commits** — Click the commit count (e.g., "5 commits") to see your full history
 - Each commit shows: the message, when it happened, and what files changed
 - Click any commit to see exactly what was added, changed, or removed
 - This is your project's timeline — every save point, in order
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Troubleshooting Prompts

Push failed:

```
I tried to push to GitHub but got an error.  
Here's what happened: [paste the error]
```

Authentication issues:

```
I'm getting an authentication error when trying to push  
to GitHub. Help me fix it.
```

Accidentally committed something wrong:

I just committed something I didn't mean to.
Can you undo the last commit but keep my files?

Not sure if Git is set up:

Check if Git is initialized in this project and if it's
connected to a GitHub remote. Show me the status.

Quick Reference

I want to...	Say this to Claude Code
Save my work	<i>"Commit the changes and push to GitHub"</i>
Get the latest from GitHub	<i>"Pull the latest changes from GitHub"</i>
Copy an existing repo	<i>"Clone the GitHub repo at [URL] into my Projects folder"</i>
Undo recent changes	<i>"Revert to the last commit"</i>
See what changed	<i>"Show me what files have changed since the last commit"</i>
See my history	<i>"Show me my recent commit history"</i>
Check Git status	<i>"What's the current Git status of this project?"</i>
Create a new branch	<i>"Create a new branch called [name]"</i>